RoMo-GS: Efficient Online 3D Gaussian Splatting for Robot Mapping via Exposure-Integrated Motion Blur

Sangwoo Jung[1,2], Di Deng[1], Jiahui Fu[1], Laura Herlant[1*], Ayoung Kim[2*], Yafei Hu[1*]

[1] RAI Institute, United States [2] Seoul National University, South Korea * Co-corresponding authors

arXiv

Personal website

Robotics and AI Institute

Motivation & Overview

Motion blur in robot mapping

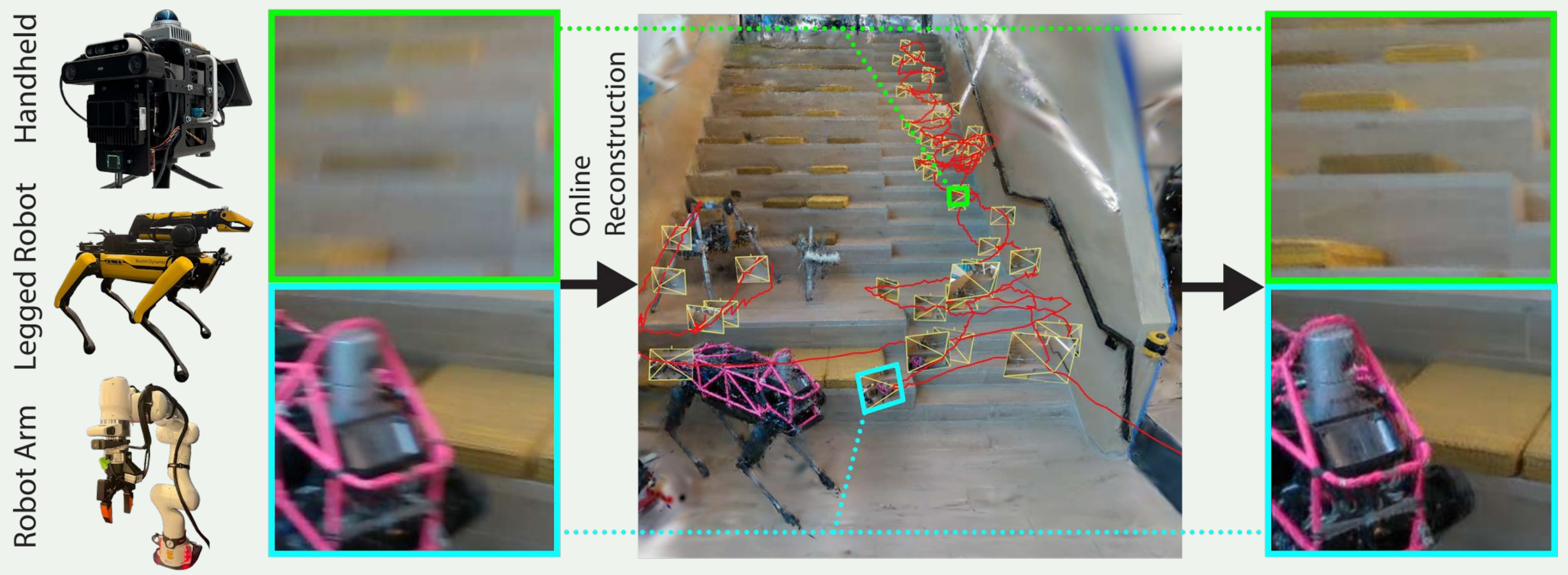
- Blurred observations degrade reconstruction during robot motion.

Computational cost of multi-sampling

- *N* rasterizations per observation, with **O(*N*)** computational cost.
- Larger *N* reduces artifacts but increases latency for online mapping.

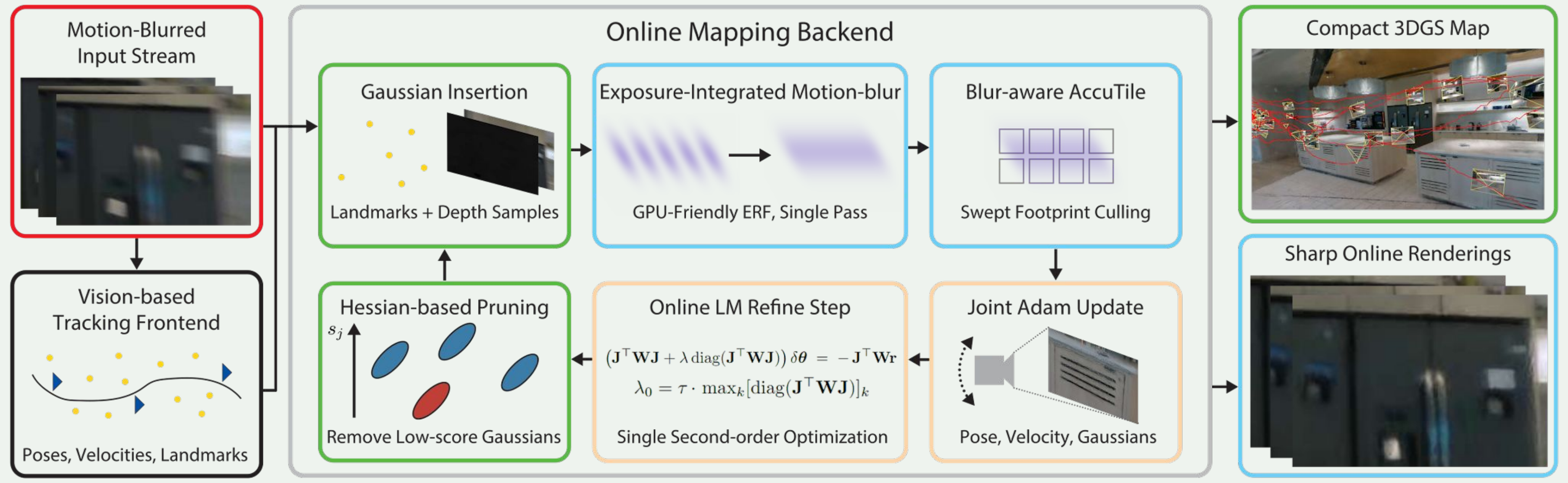
Online map growth

- Redundant Gaussians accumulate during continuous mapping.



(a) Input Blurry Images from Robots (b) Reconstructed Scene with Camera Trajectory (c) Rendered Sharp Images

Online Mapping Pipeline



Motion-Blurred Input Stream

Vision-based Tracking Frontend

Poses, Velocities, Landmarks

Online Mapping Backend

Gaussian Insertion

Landmarks + Depth Samples

Exposure-Integrated Motion-blur

GPU-Friendly ERF, Single Pass

Blur-aware AccuTile

Swept Footprint Culling

Joint Adam Update

Pose, Velocity, Gaussians

Online LM Refine Step

$$(\mathbf{J}^\top\mathbf{W}\mathbf{J} + \lambda\,\mathrm{diag}(\mathbf{J}^\top\mathbf{W}\mathbf{J}))\,\delta\boldsymbol{\theta} = -\mathbf{J}^\top\mathbf{W}\mathbf{r}$$

$$\lambda_0 = \tau \cdot \max_k[\mathrm{diag}(\mathbf{J}^\top\mathbf{W}\mathbf{J})]_k$$

Single Second-order Optimization

Hessian-based Pruning

Remove Low-score Gaussians

Compact 3DGS Map

Sharp Online Renderings

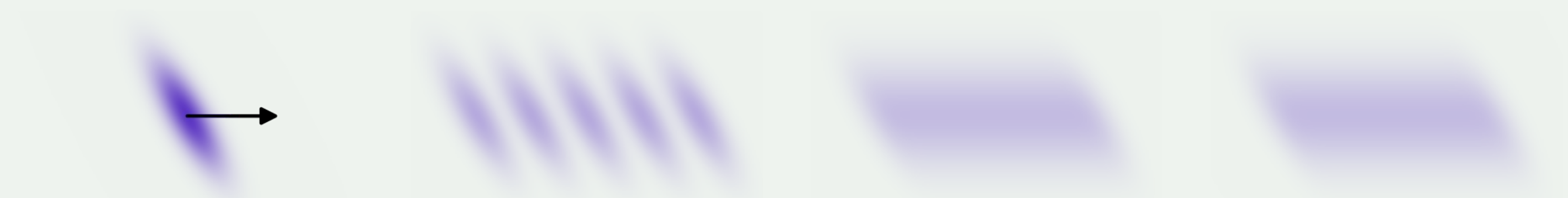
Exposure-Integrated Motion Blur

Screen-space motion

- Integrate projected opacity along a linear screen-space trajectory during the exposure.

$$\overline{\alpha}_j(x) = \frac{1}{t_e}\int_{-t_e/2}^{t_e/2} \alpha_j(x,\ \mu_j' + v_j'\tau)\,\mathrm{d}\tau$$

$\tau \in [-t_e/2,\ t_e/2]$ t_e: exposure duration



(a) Static Gaussian (b) $N_{\mathrm{blur}} = 5$ (c) $N_{\mathrm{blur}} = 100$ (d) Ours ($N_{\mathrm{blur}} = 1$)

Analytic evaluation

- Evaluate opacity with erf in one rasterization pass.
- **O(1)** cost per Gaussian, independent of sample count.

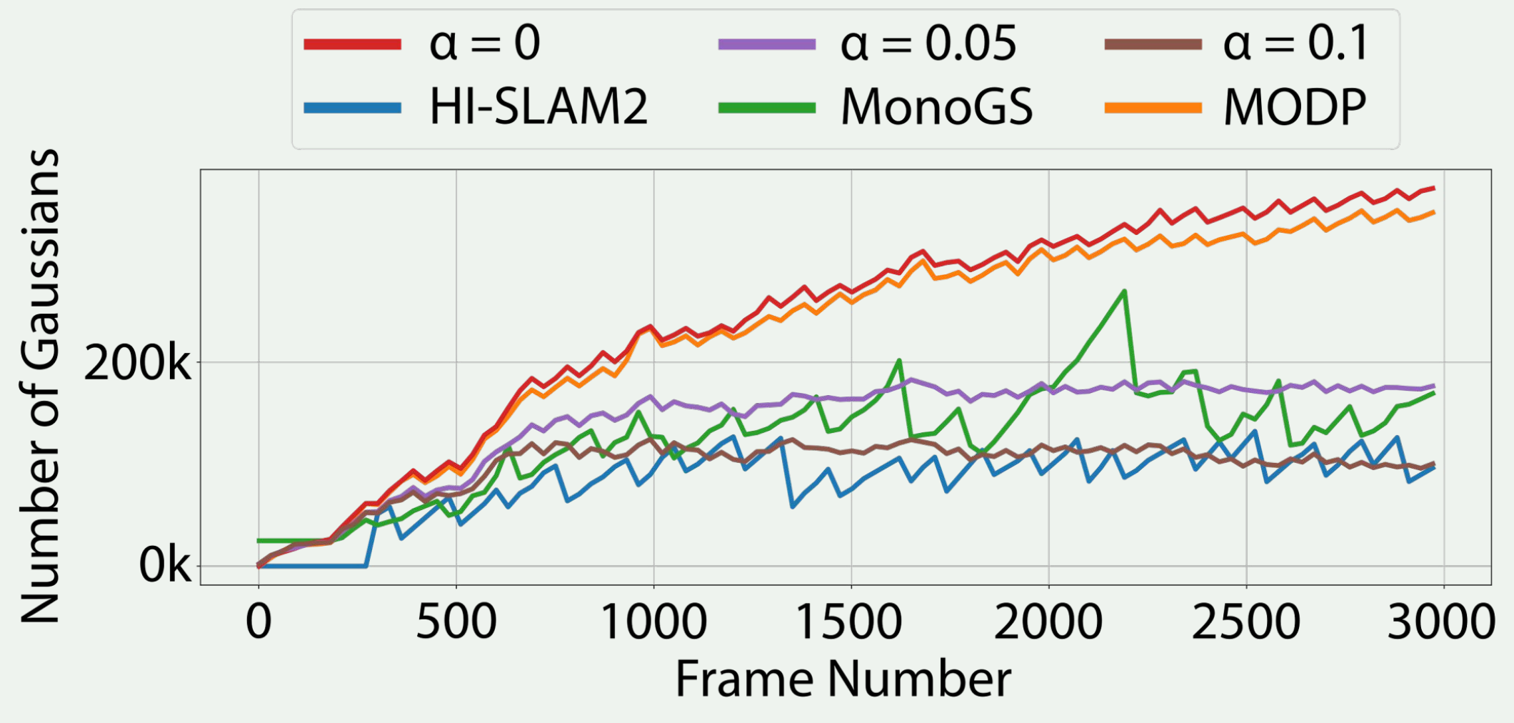
Hessian-Based Online Pruning

Opacity sensitivity

- Reuse opacity gradients from the online photometric loss.

$$s_j = \sum_i \sum_x \left(\frac{\partial L_i}{\partial o_j}\right)^2$$

- Prune low-score Gaussians observed in >10 keyframes.



dog_park: 3,000 frames, 4 loops

~20% fewer Gaussians

compared with MODP

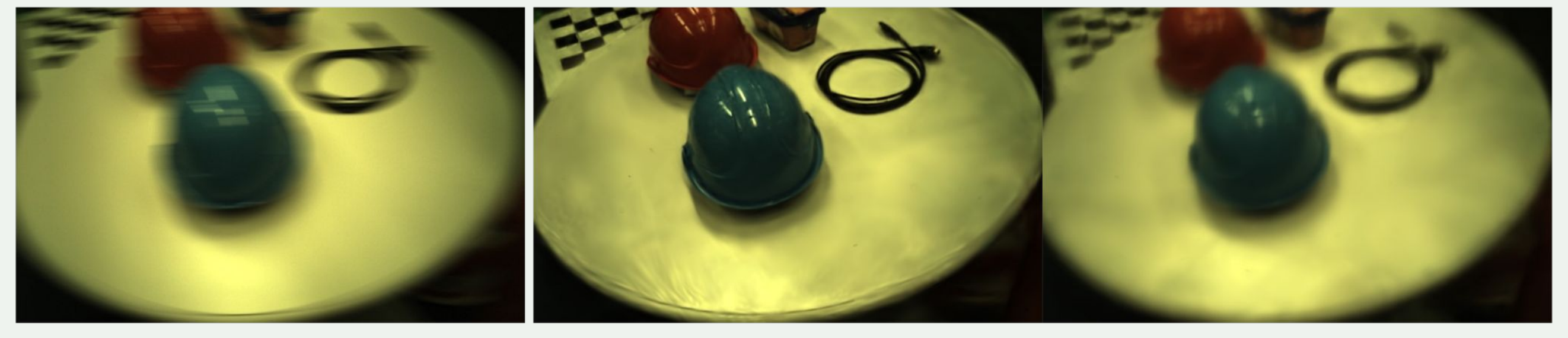
TUM: 167k vs. 210k

Custom: 309k vs. 386k

Comparable quality on TUM and custom datasets (blur model disabled).

Deblurring Quality & Efficiency

Online deblurring on ETH3D

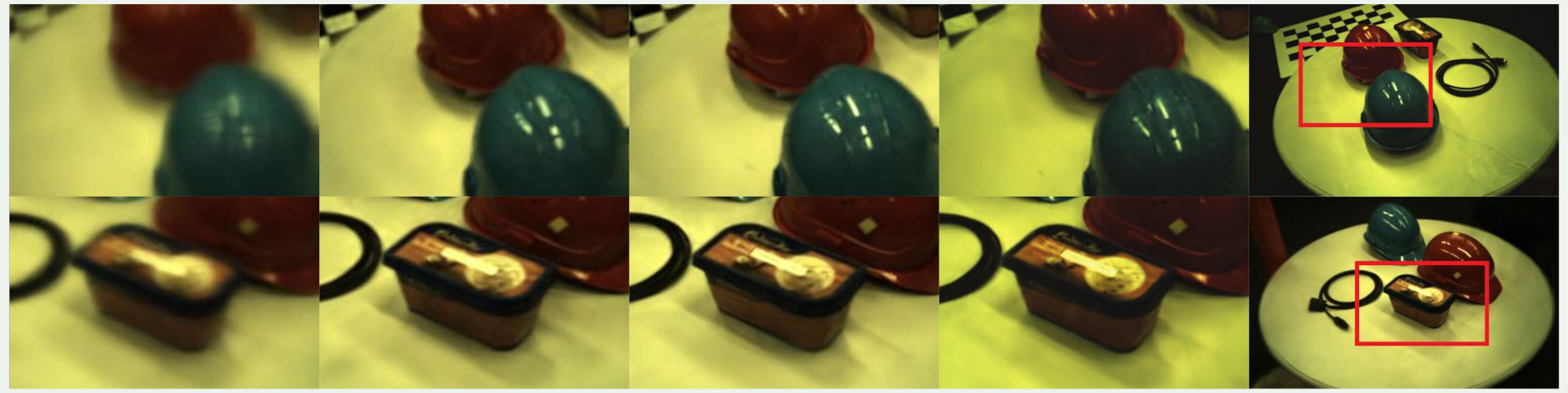


(a) Input Blurry Image (b) Ours (c) MODP

Method	PSNR ↑	LPIPS ↓	# Gaussians
MODP	23.726	0.244	129,165
RoMo-GS	**25.648**	**0.113**	**106,285**

ETH3D camera_shake_1 with BAD-Gaussians reference renderings.

Exposure integration on ETH3D



w/o Blur Reduction $N_{\mathrm{blur}} = 5$ $N_{\mathrm{blur}} = 10$ Ours Reference

Method	PSNR ↑	SSIM ↑	LPIPS ↓	FPS ↑	# Gaussians ↓
w/o blur reduction	24.355	0.892	0.202	5.698	129,128
Multi-sample, *N* = 5	25.951	0.911	0.121	2.448	124,831
Multi-sample, *N* = 10	26.003	0.911	0.107	1.592	126,902
RoMo-GS	25.367	0.905	0.113	3.695	117,669

2.32× faster than *N* = 10 multi-sampling

ETH3D comparison using the same mapping pipeline and BAD-Gaussians reference renderings.

Deblur-NeRF synthetic

~15–149× faster than offline baselines (total optimization time)

Deblurring 3DGS GSOM BAGS CoMoGaussian Ours Reference